Dr B R Ambedkar National Institute of Technology, Jalandhar
Scheme of M. Tech Programme in Information Technology from 2023 Batch onwards

ARTIFICIAL INTELLIGENCE LAB (IT-515)

Dated: 14 October,2026 **Faculty:** Dr. Simranjit Singh

Objectives:

- To understand the basic concepts of Neural Networks and Fuzzy Logic.
- To implement a Perceptron Neural Network for binary classification.
- To study the effect of learning rate and iterations on Perceptron learning and convergence.
- To implement fundamental Fuzzy Set Operations and Fuzzy Relations.

S. No.	Experiment / Program	Topic
1	Implement a Perceptron Neural Network for a simple binary classification problem. Train the Perceptron using the Perceptron Learning Algorithm and study the effect of different learning rates and numbers of iterations on its convergence and classification performance.	Neural Networks
2	Implement Fuzzy Set Operations and Fuzzy Relations. Perform union, intersection, and complement operations on fuzzy sets, and implement the composition of fuzzy relations. Demonstrate the results using suitable examples.	Fuzzy Logic